

Task 4 - Twitter Sentiment Analysis

Introduction

This project focuses on analyzing sentiment patterns in social media conversations to understand public opinion and attitudes toward various brands and topics. Using a Twitter sentiment dataset, the analysis explores positive, negative, neutral, and irrelevant sentiments expressed across different entities.

The project involves:

- data exploration
- exploratory data analysis (EDA)
- data cleaning and preprocessing
- sentiment distribution analysis
- topic and brand-level sentiment analysis
- tweet text analysis
- interactive dashboard development

The objective is to identify sentiment trends, evaluate public perception of brands and topics, and uncover meaningful insights from social media discussions.

Python was used for data cleaning, preprocessing, and exploratory analysis, while Tableau was utilized to create interactive visualizations and dashboards for effective storytelling.

Dataset Source

Dataset: Twitter Entity Sentiment Analysis Dataset

Source: Kaggle – Twitter Entity Sentiment Analysis

The dataset contains social media posts associated with various brands, products, organizations, and topics. Each tweet is labeled with a sentiment category, enabling analysis of public opinion and sentiment patterns across different entities.

Project Objectives

- Understand the structure and characteristics of the social media dataset
- Perform exploratory data analysis to identify sentiment patterns
- Clean and preprocess tweet data for analysis
- Analyze the distribution of positive, negative, neutral, and irrelevant sentiments
- Identify the most discussed brands and topics
- Compare sentiment trends across different entities
- Explore public attitudes and opinions toward specific brands and topics
- Create an interactive Tableau dashboard to communicate insights effectively

Workflow

1. Import Libraries
2. Load Training & Validation Datasets
3. Merge Datasets
4. Basic Data Exploration
5. Data Cleaning and Missing Value Handling
6. Exploratory Data Analysis (EDA)
7. Sentiment Distribution Analysis
8. Top Topics/Brands Analysis
9. Topic-wise Sentiment Analysis
10. Export Clean Dataset
11. Sentiment Comparison Across Topics
12. Data Preparation for Visualization
13. Tableau Dashboard Development
14. Insight Generation
15. Conclusion

Tools & Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Tableau
- Jupyter Notebook

1. Import Libraries

In [5]:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

print("All Library's Imported")
```

All Library's Imported

Import Dataset

In [6]:

```
train = pd.read_csv(
    "D:/Tanushree/prodigy/Task 4/twitter/twitter_training.csv",
    header=None,
    names=["ID", "Topic", "Sentiment", "Tweet"]
)
val = pd.read_csv(
    "D:/Tanushree/prodigy/Task 4/twitter/twitter_validation.csv",
    header=None,
    names=["ID", "Topic", "Sentiment", "Tweet"]
)
```

In [7]:

```
train.shape
```

Out[7]:

```
(74683, 4)
```

In [8]:

```
val.shape
```

Out[8]:

```
(1000, 4)
```

In [9]:

```
df = pd.concat([train, val], ignore_index=True)

print(df.shape)
```

```
(75683, 4)
```

In [10]:

```
df.columns=["ID", "Topic", "Sentiment", "Tweet"]

print(df.head())
print(df.info())
```

```

      ID      Topic Sentiment \
0  Column1  Column2  Column3
1    2401  Borderlands  Positive
2    2401  Borderlands  Positive
3    2401  Borderlands  Positive
4    2401  Borderlands  Positive

                                     Tweet
0                                     Column4
1  im getting on borderlands and i will murder yo...
```

```

2 I am coming to the borders and I will kill you...
3 im getting on borderlands and i will kill you ...
4 im coming on borderlands and i will murder you...
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 75683 entries, 0 to 75682
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0   ID           75683 non-null  object
1   Topic        75683 non-null  object
2   Sentiment    75683 non-null  object
3   Tweet        74997 non-null  object
dtypes: object(4)
memory usage: 2.3+ MB
None

```

In [11]:

```

df.columns=["ID","Topic","Sentiment","Tweet"]

# Remove the first row containing Column1, Column2...
df = df.iloc[1:].reset_index(drop=True)

print(df.head())

```

	ID	Topic	Sentiment	Tweet
0	2401	Borderlands	Positive	
1	2401	Borderlands	Positive	
2	2401	Borderlands	Positive	
3	2401	Borderlands	Positive	
4	2401	Borderlands	Positive	


```

0 im getting on borderlands and i will murder yo...
1 I am coming to the borders and I will kill you...
2 im getting on borderlands and i will kill you ...
3 im coming on borderlands and i will murder you...
4 im getting on borderlands 2 and i will murder ...

```

Basic EDA

In [12]:

```

# Missing/null

df.isnull().sum()

```

Out[12]:

```

ID           0
Topic        0
Sentiment    0
Tweet       686
dtype: int64

```

In [13]:

```

# Dropping Null

df = df.dropna(subset=["Tweet"])

df.shape

```

```
Out[13]:  
(74996, 4)
```

```
In [14]:  
df.isnull().sum()
```

```
Out[14]:  
ID            0  
Topic         0  
Sentiment     0  
Tweet         0  
dtype: int64
```

Sentiment Distribution Analysis

```
In [15]:  
df["Sentiment"].value_counts()
```

```
Out[15]:  
Sentiment  
Negative      22624  
Positive      20932  
Neutral       18393  
Irrelevant    13047  
Name: count, dtype: int64
```

```
In [16]:  
round(df["Sentiment"].value_counts(normalize=True)*100,2)
```

```
Out[16]:  
Sentiment  
Negative      30.17  
Positive      27.91  
Neutral       24.53  
Irrelevant    17.40  
Name: proportion, dtype: float64
```

Plot: Sentiment Distribution (Must Have)

```
In [26]:  
plt.figure(figsize=(8,5))  
  
ax = sns.countplot(  
    data=df,  
    x="Sentiment",  
    hue="Sentiment",  
    palette="Dark2",  
    legend=False  
)  
  
for p in ax.patches:  
    height = p.get_height()  
    if height > 0:  
        ax.annotate(  
            f'{int(height)}',  
            (p.get_x() + p.get_width() / 2., height),  
            ha='center',
```

```

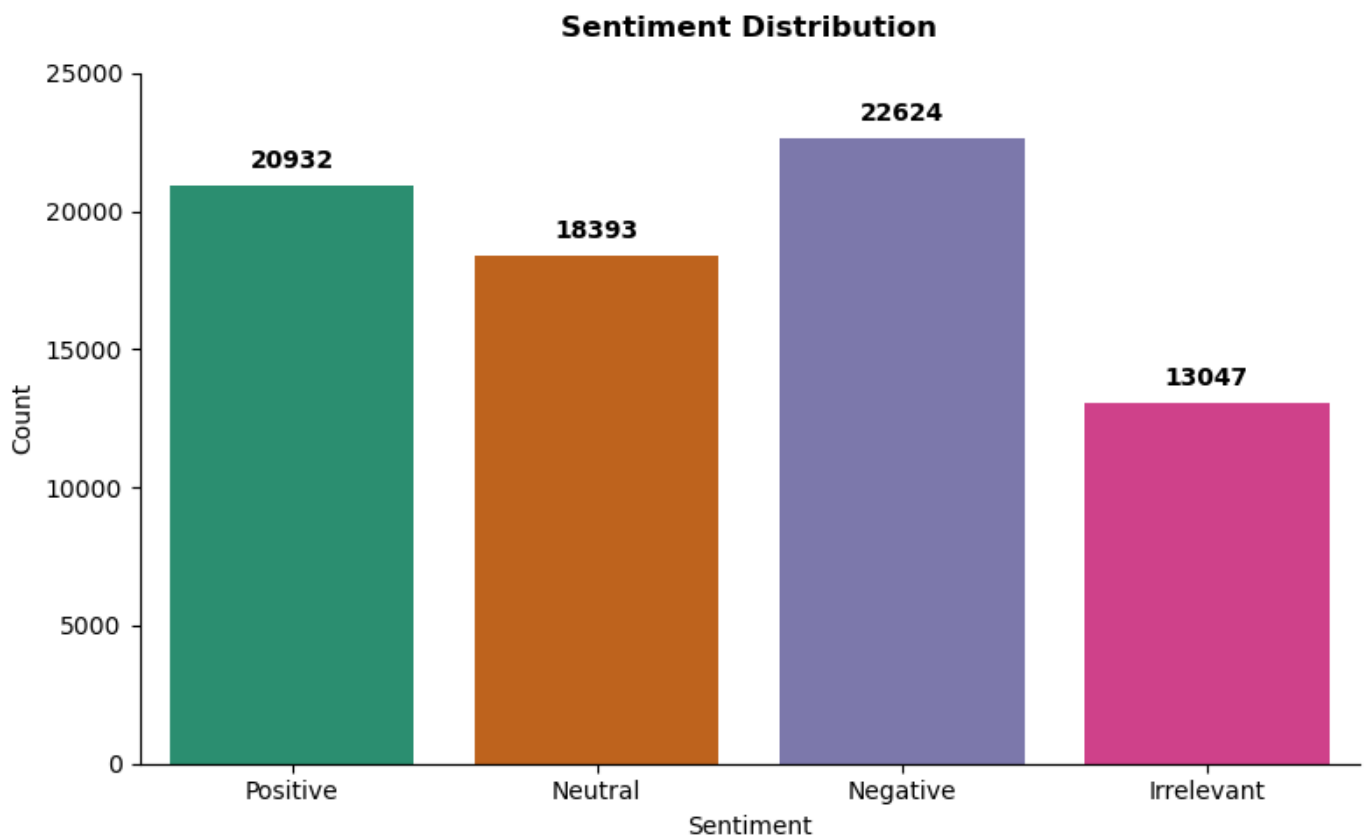
        va='bottom',
        fontsize=10,
        fontweight='bold',
        xytext=(0, 5),
        textcoords='offset points'
    )

plt.yticks([0, 5000, 10000, 15000, 20000, 25000])

plt.title("Sentiment Distribution", fontsize=12, fontweight='bold', pad=15)
plt.xlabel("Sentiment")
plt.ylabel("Count")

sns.despine(top=True, right=True)
plt.tight_layout()
plt.show()

```



Top Topics/Brands Analysis

```

In [43]:
top_topics = df["Topic"].value_counts().head(15)

top_topics_df = pd.DataFrame({
    "Topic": top_topics.index,
    "Tweet Count": top_topics.values
})

print(type(top_topics_df))
print(top_topics_df.head())

```

```
<class 'pandas.core.frame.DataFrame'>
```

	Topic	Tweet Count
0	LeagueOfLegends	2414
1	CallOfDuty	2407
2	MaddenNFL	2406
3	TomClancysRainbowSix	2399
4	Verizon	2397

Plot Top 10 Topics/Brands

In [47]:

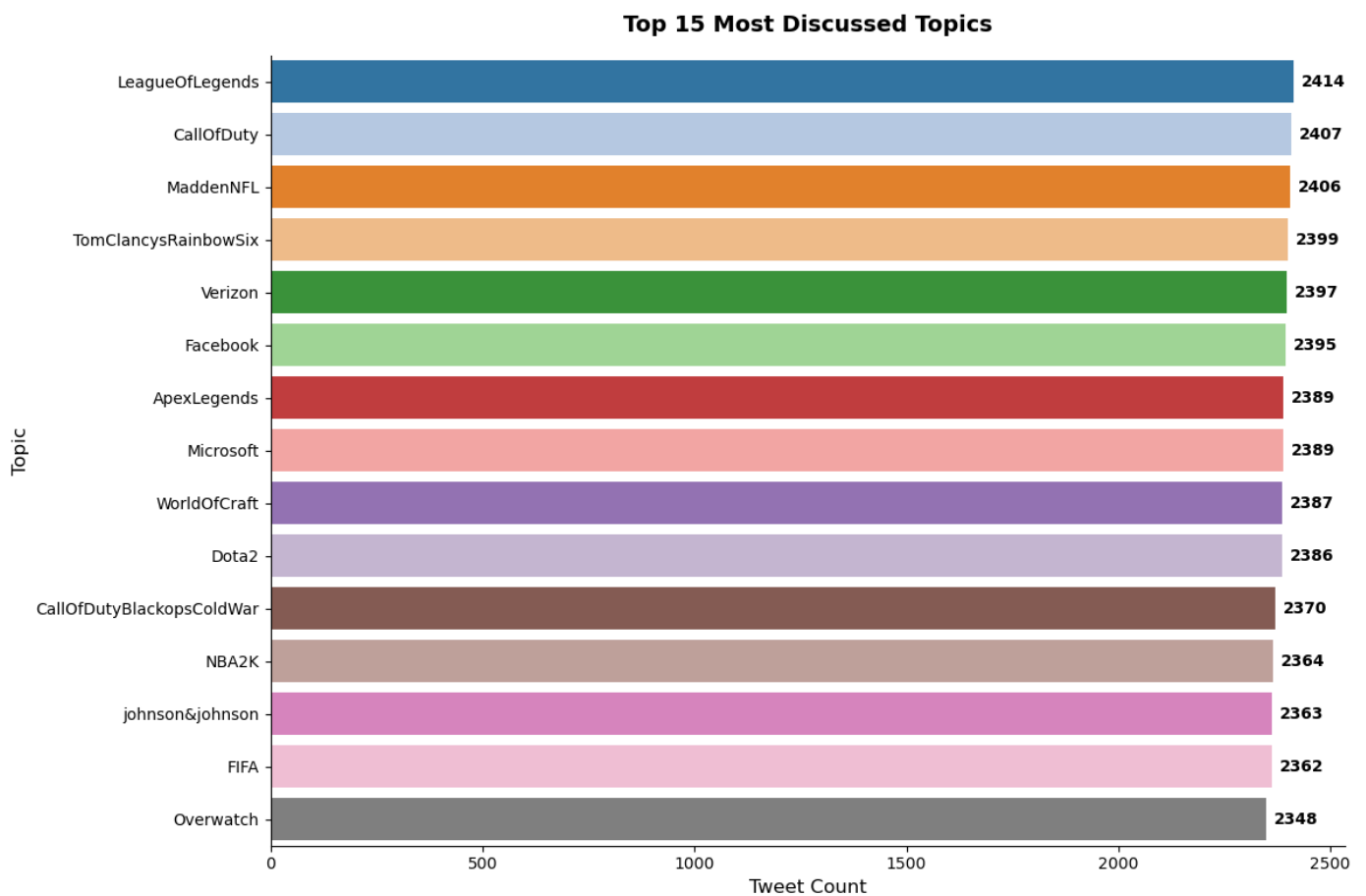
```
plt.figure(figsize=(12,8))

ax = sns.barplot(
    data=top_topics_df,
    x="Tweet Count",
    y="Topic",
    hue="Topic",
    palette="tab20",
    legend=False
)

for p in ax.patches:
    width = p.get_width()
    if width > 0:
        ax.annotate(
            f'{int(width)}',
            (width, p.get_y() + p.get_height() / 2.),
            ha='left',
            va='center',
            fontsize=10,
            fontweight='bold',
            xytext=(5, 0),
            textcoords='offset points'
        )

sns.despine(top=True, right=True)
plt.title("Top 15 Most Discussed Topics", fontsize=14, fontweight='bold', pad=15)
plt.xlabel("Tweet Count", fontsize=12)
plt.ylabel("Topic", fontsize=12)

plt.tight_layout()
plt.show()
```



Sentiment by Topics

In [19]:

```
topic_sentiment = pd.crosstab(
    df["Topic"],
    df["Sentiment"]
)

topic_sentiment.head()
```

Out[19]:

Sentiment	Irrelevant	Negative	Neutral	Positive
Topic				
Amazon	189	581	1225	315
ApexLegends	195	597	953	644
AssassinsCreed	265	379	160	1463
Battlefield	927	471	355	589
Borderlands	242	431	609	1031

Plot: Top 10 topics

In [53]:

```

top_topics = df["Topic"].value_counts().head(10).index

filtered = df[df["Topic"].isin(top_topics)]

ct = pd.crosstab(filtered["Topic"], filtered["Sentiment"])

sentiment_colors = {
    'Positive': '#4CAF50',
    'Negative': '#FF3B30',
    'Neutral': '#FF9500',
    'Irrelevant': '#007BFF'
}

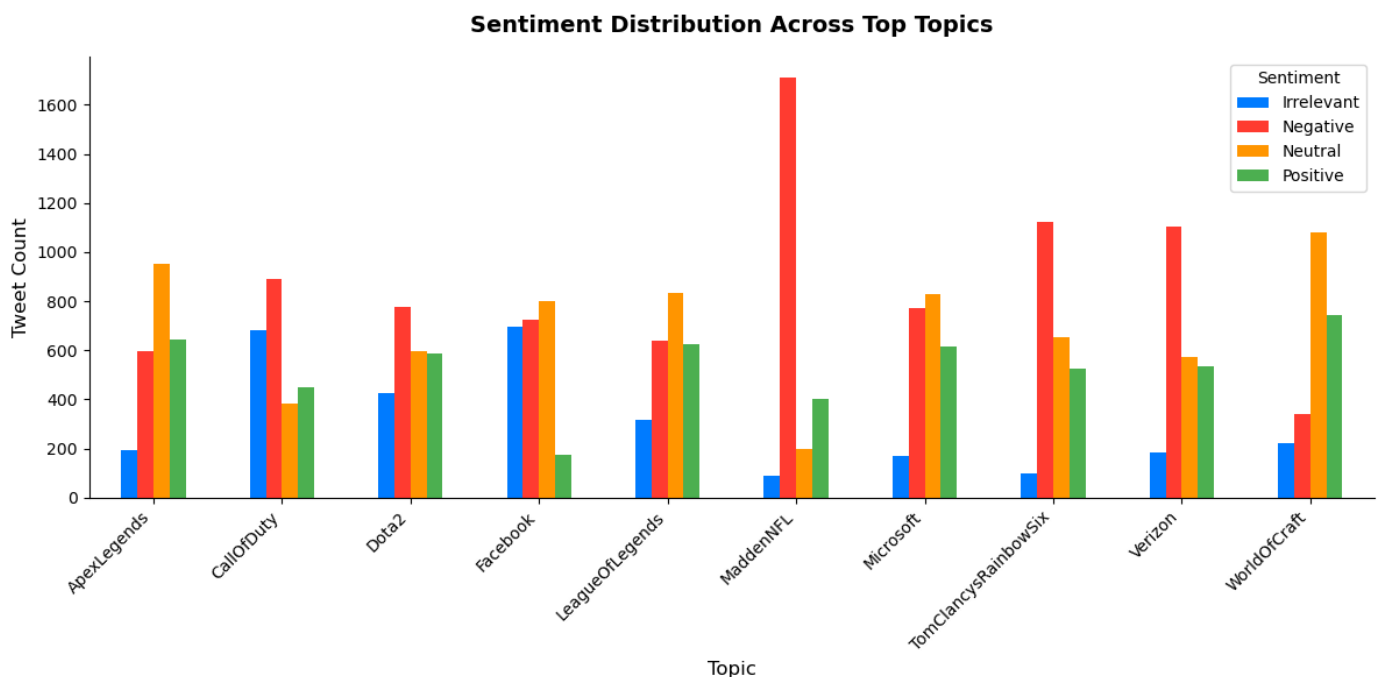
ax = ct.plot(
    kind="bar",
    figsize=(12, 6),
    color=sentiment_colors
)

plt.title("Sentiment Distribution Across Top Topics", fontsize=14, fontweight='bold', pa
plt.xlabel("Topic", fontsize=12)
plt.ylabel("Tweet Count", fontsize=12)
plt.xticks(rotation=45, ha='right')

import seaborn as sns
sns.despine(top=True, right=True)

plt.tight_layout()
plt.show()

```



Export Clean Dataset

In [20]:

```
import os

print(os.getcwd())
```

C:\Users\tanushree

In [21]:

```
df.to_csv(
    "twitter_cleaned.csv",
    index=False
)
```

In [22]:

```
df.to_csv(
    r"D:\Tanushree\prodigy\Task 4\twitter\twitter_cleaned.csv",
    index=False
)
```

In [23]:

```
os.listdir(r"D:\Tanushree\prodigy\Task 4\twitter")
```

Out[23]:

```
['twitter_cleaned.csv', 'twitter_training.csv', 'twitter_validation.csv']
```

Conclusion/Key Insights

- Social media sentiment is strongly influenced by brand and topic engagement patterns.
- Tech and gaming brands generally receive more positive public sentiment.
- Certain brands and topics show concentrated negative sentiment, indicating customer concerns.
- Clear differences in sentiment highlight opportunities for improving brand perception and user experience.